The documentation and process conversion measures necessary to comply with this revision shall be completed by 14 April 2002.

INCH-POUND

MIL-PRF-19500/558D
14 January 2002
SUPERSEDING
MIL-PRF-19500/558C
31 August 1998

PERFORMANCE SPECIFICATION

SEMICONDUCTOR DEVICE, FOUR TRANSISTOR ARRAY,UNITIZED,
PNP, SILICON, SWITCHING TYPES 2N6987, 2N6987U, AND 2N6988,
JAN, JANTX, JANTXV, AND JANS

This specification is approved for use by all Departments
and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for PNP, silicon, switching transistors, four independent chip array. Four levels of product assurance are provided for each device type as specified in MIL-PRF-19500.

1.2 Physical dimensions. See figures 1, 2, 3, and 4 (14-pin dual-in-line, 14-pin flat-pack, and 20-pin leadless chip carrier) and 3.4.

1.3 Maximum ratings. (1)

| | $P_T$ $T_A$ = +25°C (2) | $V_{CBO}$ (3) | $V_{EBO}$ (3) | $V_{CEO}$ (3) | $I_C$ (3) | $T_{OP}$ and $T_{STG}$ |
|---|---|---|---|---|---|---|
| | W | V dc | V dc | V dc | mA dc | °C |
| 2N6987 | 1.5 | 60 | 5 | 60 | 600 | -65 to +200 |
| 2N6987U | 1.0 | 60 | 5 | 60 | 600 | -65 to +200 |
| 2N6988 | 0.4 | 60 | 5 | 60 | 600 | -65 to +200 |

(1) Maximum voltage between transistors shall be ≥ 500 V dc.

* (2) Derate linearly 8.57 mW/°C above $T_A$ = +25°C for 2N6987 and 5.71 mW/°C for 2N6987U. Derate linearly 2.286 mW/°C above $T_A$ = +25°C for 2N6988. Ratings apply to total package.

(3) Ratings apply to each transistor in the array.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Supply Center, Columbus, ATTN: DSCC-VAC, P. O. Box 3990, Columbus, OH 43216- 5000, by using Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of the document or by letter.

AMSC N/A FSC 5961

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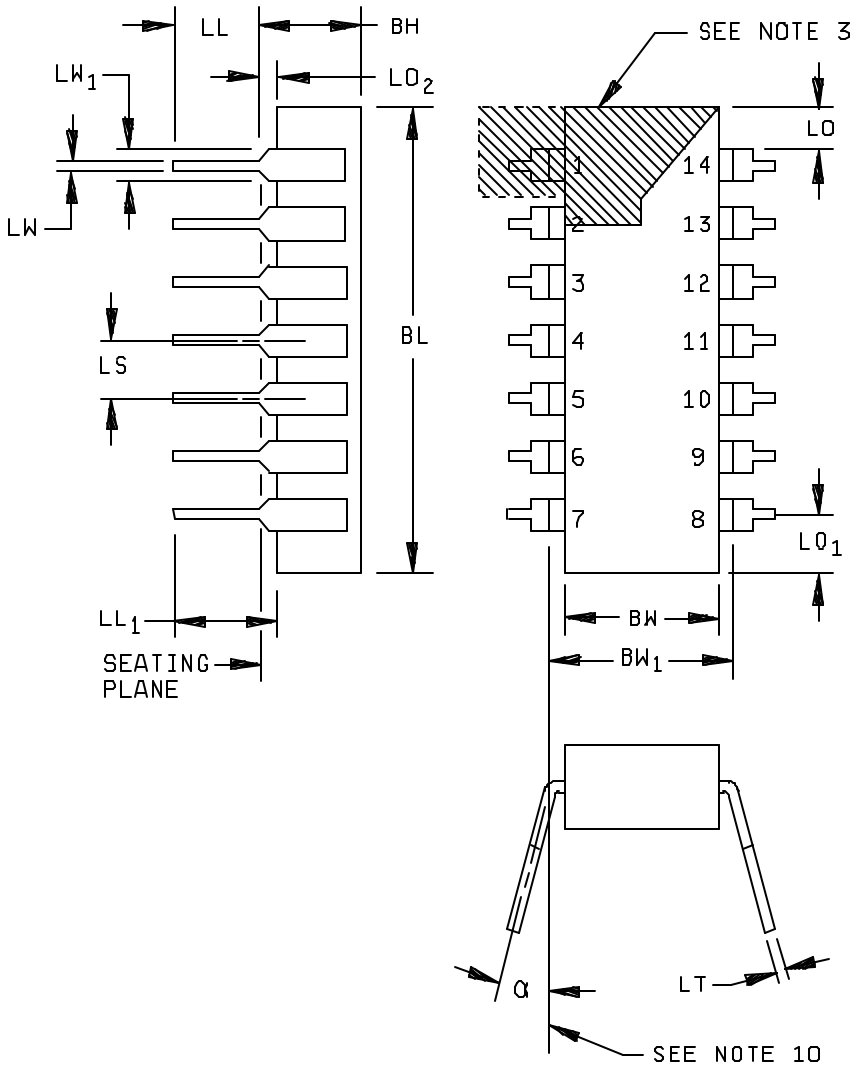



FIGURE 1. Dimensions and configuration for type 2N6987.



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| Symbol | Inches | | Millimeters | | Notes |
|---|---|---|---|---|---|
| | Min | Max | Min | Max | |
| BH | | .200 | | 5.08 | |
| LW | .014 | .023 | 0.36 | 0.58 | 8 |
| $LW_1$ | .030 | .070 | 0.76 | 1.78 | 4, 8 |
| LT | .008 | .015 | 0.20 | 0.38 | 8 |
| BL | | .785 | | 19.94 | 4 |
| BW | .220 | .310 | 5.59 | 7.87 | 4 |
| $BW_1$ | .290 | .320 | 7.37 | 8.13 | 7 |
| LS | .100 BSC | | 2.54 BSC | | 5, 9 |
| LL | .125 | .200 | 3.18 | 5.08 | |
| $LL_1$ | .150 | | 3.81 | | |
| $LO_2$ | .015 | .060 | 0.38 | 1.52 | 3 |
| $LO_1$ | | .098 | | 2.49 | 6 |
| LO | .005 | | 0.13 | | 6 |
| $\alpha$ | 0° | 15° | 0° | 15° | |

NOTES:

1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Index area; a notch or pin one identification mark shall be located adjacent to pin one and shall be located within the shaded area shown. The manufacturer's identification shall not be used as a pin one identification mark.
4. The minimum limit for dimension $LW_1$ may be .023 inch (0.58 mm) for lead numbers 1, 7, 8, and 14 only.
5. Dimension $LO_2$ shall be measured from the seating plane to the base plane.
6. This dimension allows for off-center lid, meniscus, and glass overrun.
7. The basic pin spacing is .100 inch (2.54 mm) between centerlines. Each pin centerline shall be located within ±.010 inch (± 0.25 mm) of its exact longitudinal position relative to pins 1 and 14.
8. Applies to all four corners (lead numbers 1, 7, 8, and 14).
9. Lead center when $\alpha$ is 0. $BW_1$ shall be measured at the centerline of the leads.
10. All leads: Increase maximum limit by .003 inch (0.08 mm) measured at the center of the flat, when lead finish A is applied. Pointed or round lead ends are allowed.
11. Twelve spaces.
12. No organic or polymeric materials shall be molded to the bottom of the package to cover leads.
13. For terminal connections see figure 4.

FIGURE 1. Dimensions and configuration for type 2N6987 - Continued.



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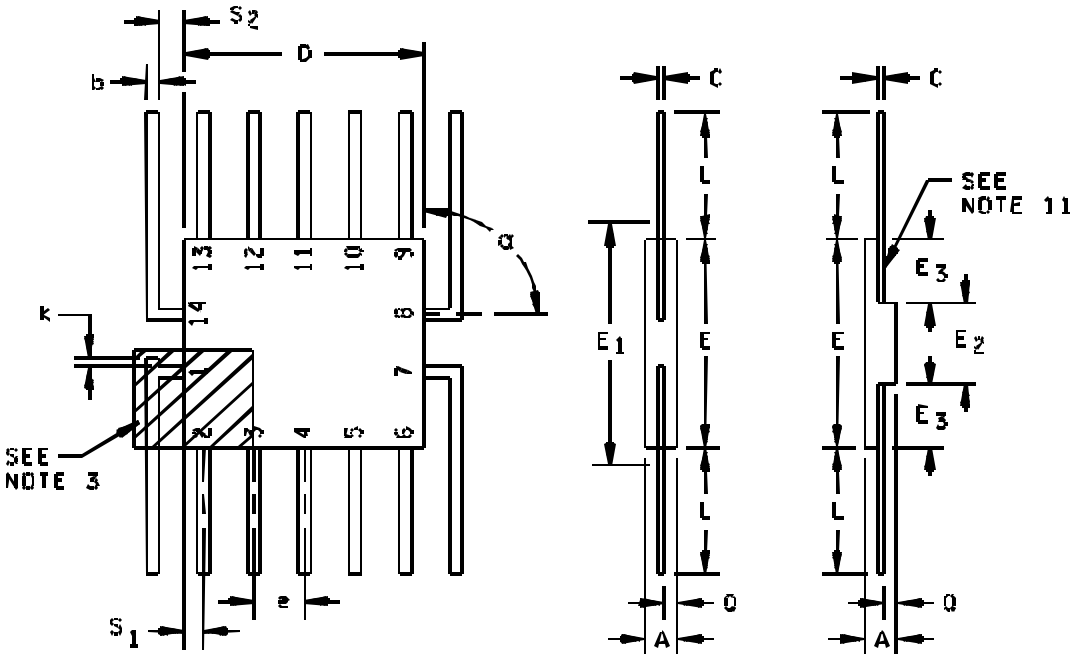



FIGURE 2. Physical dimensions for type 2N6988.



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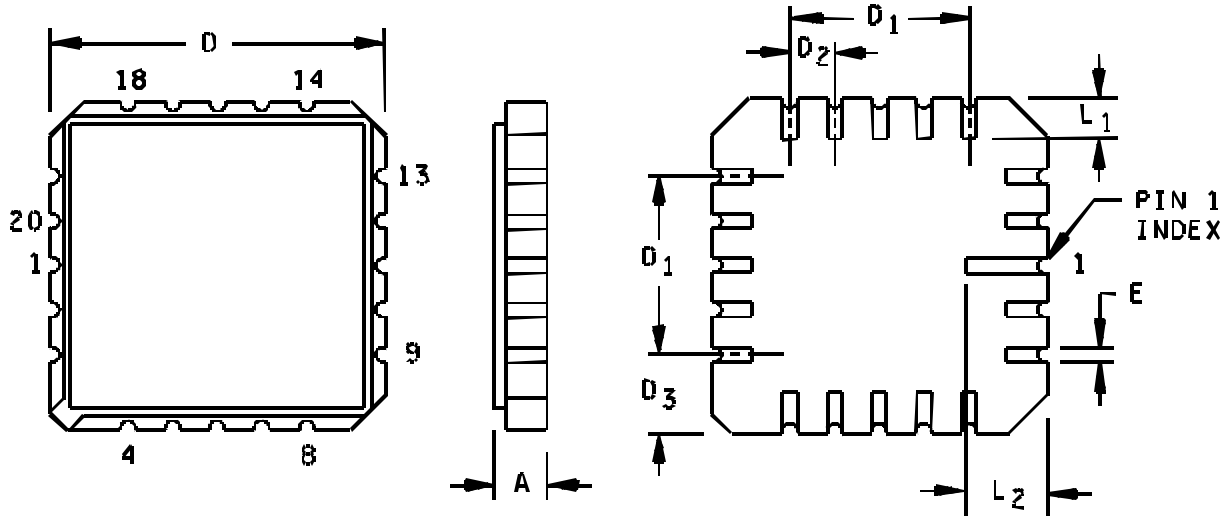
| Symbol | Dimensions | | | | Notes |
|---|---|---|---|---|---|
| | Inches | | Millimeters | | |
| | Min | Max | Min | Max | |
| A | .030 | .115 | 0.76 | 2.92 | |
| b | .010 | .019 | 0.25 | 0.48 | 7 |
| C | .003 | .006 | 0.08 | 0.15 | 7 |
| D | | .280 | | 7.11 | 5 |
| E | .240 | .260 | 6.10 | 6.60 | |
| $E_1$ | | .290 | | 7.37 | 5 |
| $E_2$ | .125 | | 3.18 | | |
| $E_3$ | .030 | | 0.76 | | |
| e | 050 BSC | | 1.27 BSC | | 6, 8 |
| k | .008 | .015 | 0.20 | 0.38 | 12 |
| L | .250 | .370 | 6.35 | 9.40 | |
| Q | .005 | .040 | 0.13 | 1.02 | 4 |
| $S_1$ | .005 | | 0.13 | | 9, 10 |
| $S_2$ | .004 | | 0.10 | | 13 |
| α | 30° | 90° | 30° | 90° | 14 |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Index area; a notch or pin one identification mark shall be located adjacent to pin one and shall be located within the shaded area shown. The manufacturer's identification shall not be used as a pin one identification mark. Alternatively, a tab (dim k) may be used to identify pin one.
4. Dimension Q shall be measured at the point of exit of the lead from the body.
5. This dimension allows for off-center lid, meniscus and glass overrun.
6. The basic pin spacing is .050 inch (1.25 mm) between centerlines. Each pin centerline shall be located within ± .005 inch (±0.013 mm) of its exact longitudinal position relative to pins 1 and 14.
7. All leads - increase maximum limit by .003 inch (0.08 mm) measured at the center of the flat when the lead finish is solder.
8. Twelve spaces.
9. Applies to all four corners (leads number 2, 6, 9, and 13).
10. Dimension $S_1$ may be .000 inch (0.00 mm) if leads number 2, 6, 9, and 13 bend toward the cavity of the package within one lead width from the point of entry of the lead into the body or if the leads are brazed to the metallized ceramic body.
11. No organic or polymeric materials shall be molded to the bottom of the package to cover the leads.
12. Optional, see note 1. If a pin one identification mark is used in addition to this tab, the minimum limit of dimension k does not apply.
13. Applies to leads number 1, 7, 8, and 14.
14. Lead configuration is optional within dimension E, except dimensions b and C apply.

FIGURE 2. Physical dimensions for type 2N6988 - Continued.



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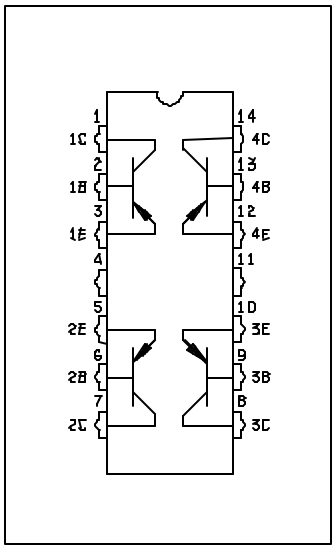

| Symbol | Dimensions | | | |
|---|---|---|---|---|
| | Inches | | Millimeters | |
| | Min | Max | Min | Max |
| A | .063 | .075 | 1.60 | 1.90 |
| D | .345 | .355 | 8.76 | 9.02 |
| $D_1$ | .195 | .205 | 4.95 | 5.21 |
| $D_2$ | .050 TYP | | 1.27 TYP | |
| $D_3$ | .070 | .080 | 1.76 | 2.03 |
| E | .025 REF | | 0.64 REF | |
| $L_1$ | .050 REF | | 1.27 REF | |
| $L_2$ | .080 | .090 | 2.03 | 2.28 |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ± .005 inch (0.13 mm).
4. For terminal connections see figure 4.

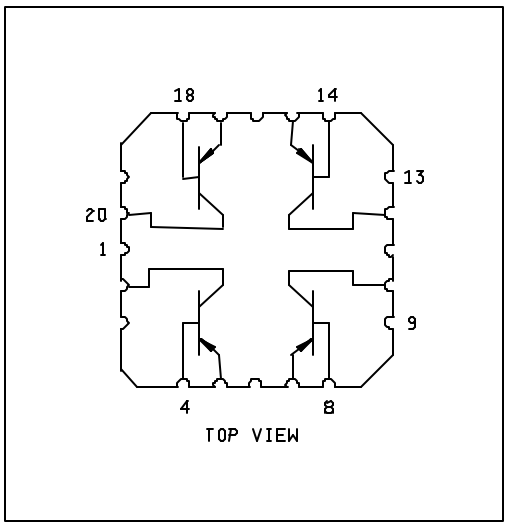
FIGURE 3. Physical dimensions for type 2N6987U.



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14-lead flat-package or dual-in-line (top view)




20 pin leadless chip carrier (top view).

FIGURE 4. Schematic and terminal connections.



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1.4 Primary electrical characteristics. Characteristics apply to each transistor in the array.

| Limits | $h_{FE2}$ (1)<br>$V_{CE}$ = 10 V dc<br>$I_C$ = 1.0 mA dc | $h_{FE4}$ (1)<br>$V_{CE}$ = 10 V dc<br>$I_C$ = 150 mA dc | $C_{obo}$<br>$V_{CB}$ = 10 V dc<br>$I_E$ = 0<br>100 kHz ≤ f ≤ 1 MHz | Switching | |
|---|---|---|---|---|---|
| | | | | $t_{on}$<br>see figure 5 | $t_{off}$<br>see figure 6 |
| | | | pF | ns | ns |
| Min | 100 | 100 | | | |
| Max | 450 | 300 | 8 | 45 | 300 |

| Limits | $h_{fe}$<br>$V_{CE}$ = 20 V dc<br>$I_C$ = 50 mA dc<br>f = 100 MHz | $V_{CE(sat)2}$ (1)<br>$I_C$ = 500 mA dc<br>$I_B$ = 50 mA dc | $V_{BE(sat)2}$ (1)<br>$I_C$ = 500 mA dc<br>$I_B$ = 50 mA dc |
|---|---|---|---|
| | | V dc | V dc |
| Min | 2.0 | | |
| Max | 8.0 | 1.6 | 2.6 |

(1) Pulsed (see 4.5.1).

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in section 3 and 4 of this specification, whether or not they are listed.



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2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

DEPARTMENT OF DEFENSE

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

STANDARD

DEPARTMENT OF DEFENSE

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Document Automation and Production Services (DAPS), Building 4D (DPM-DODSSP), 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1. General. The requirements for acquiring the product described herein shall consist of this document and MIL-PRF-19500.

3.2. Qualification. Devices furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturer's list (QML) before contract award (see 4.2 and 6.3).

3.3. Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500.

3.4 Interface requirements and physical dimensions. The interface requirements and physical dimensions shall be as specified in MIL-PRF-19500 and on figures 1 (MO-036AB), 2, 3 and 4 herein.

3.4.1 Lead finish. Lead finish shall be solderable as defined in MIL-STD-750, MIL-PRF-19500, and herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.4.2 Schematic and terminal connections. The schematic and terminal connections shall be as shown on figure 4.

3.5 Marking. Marking shall be in accordance with MIL-PRF-19500.

3.6 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3, 1.4, and table I herein.



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3.7 Electrical test requirements. The electrical test requirements shall be the subgroups specified in 4.4.2 and 4.4.3 herein.

3.8. Workmanship. Semiconductor devices shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3).

c. Conformance inspection (see 4.4).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500 and table II herein.

* 4.3 Screening (JANS, JANTX and JANTXV levels only). Screening shall be in accordance with table IV of MIL-PRF-19500 appendix E, and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

| Screen (see appendix E, table IV of MIL-PRF-19500) | Measurements | |
|---|---|---|
| | JANS level | JANTX & JANTXV levels |
| 3c | Thermal impedance, method 3131 of MIL-STD-750. | Thermal impedance, method 3131 of MIL-STD-750. |
| 9 | $I_{CBO2}$ and $h_{FE4}$ | Not applicable |
| 10 | 24 hours minimum | 24 hours minimum |
| 7(a) or 14(a) | Maximum leak rate = 1 x $10^{-7}$ atm cc/s Test condition C | Maximum leak rate = 1 x $10^{-7}$ atm cc/s Test condition C |
| 11 | $I_{CBO2}$ and $h_{FE4}$<br>$\Delta I_{CBO2}$ = 100 percent of initial value or 5 nA dc, whichever is greater;<br>$\Delta h_{FE4}$ = ±15 percent of initial value. | $I_{CBO2}$ and $h_{FE4}$ |
| 12 | See 4.3.1, 240 hours minimum | See 4.3.1, 80 hours minimum |
| 13 | Subgroups 2 and 3 of table I herein;<br>$\Delta I_{CBO2}$ = 100 percent of initial value or 5 nA dc, whichever is greater;<br>$\Delta h_{FE4}$ = ±15 percent of initial value. | Subgroup 2 of table I herein;<br>$\Delta I_{CBO2}$ = 100 percent of initial value or 5 nA dc, whichever is greater;<br>$\Delta h_{FE4}$ = ±15 percent of initial value. |



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4.3.1 Power burn-in conditions. Power burn-in conditions are as follows: $T_A$ = Room ambient as defined in the general requirements of MIL-STD-750, (see 4.5); $V_{CB}$ = 10 - 30 V dc; $P_T$ = 1.5 W for 2N6987; $P_T$ = 1.0 W for 2N6987U, and $P_T$ = 0.4 W for 2N6988. NOTE: No heat sink or forced air cooling on the devices shall be permitted. Power ratings apply to total package.

4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500 and as specified herein. If alternate screening is being performed in accordance with MIL-PRF-19500, a sample of screened devices shall be submitted to and pass the requirements of group A1 and group A2 inspection only (table VIb, group B, subgroup 1 is not required to be performed again if group B has already been satisfied in accordance with 4.4.2 herein).

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-PRF-19500 and table I herein.

* 4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table VIa (JANS) of MIL-PRF-19500 and 4.4.2.1 herein. Electrical measurements (end-points) and delta requirements shall be in accordance with group A, subgroup 2 and 4.5.5 herein, delta parameters apply to subgroups B4 and B5. See 4.4.2.2 for JAN, JANTX, and JANTXV group B testing. Electrical measurements (end-points) and delta requirements for JAN, JANTX, and JANTXV shall be after each step in 4.4.2.2 and shall be in accordance with group A, subgroup 2 and 4.5.5 herein.

* 4.4.2.1 Group B inspection, table VIa (JANS) of MIL-PRF-19500.

| Subgroup | Method | Condition |
|---|---|---|
| B4 | 1037 | $V_{CB}$ = 10 - 30 V dc; (see 4.5), 2,000 cycles. No heat sink or forced-air cooling on the devices shall be permitted. |
| B5 | 1027 | (Note: If a failure occurs, resubmission shall be at the test conditions of the original sample.) $V_{CB}$ = 10 V dc, $P_D \geq$ 100 percent of max rated $P_T$ (see 1.3).<br><br>Option 1: 96 hours minimum sample size in accordance with MIL-PRF-19500, table Via, adjust $T_A$ or $P_D$ to achieve $T_J$ = +275°C minimum.<br><br>Option 2: 216 hours minimum, sample size = 45, c = 0; adjust $T_A$ or $P_D$ to achieve $T_J$ = +225°C minimum. |



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4.4.2.2 Group B inspection (JAN, JANTX, and JANTXV). 1/

| Step | Method | Condition |
|---|---|---|
| 1 | 1039 | Steady-state life: Test condition B, 340 hours, $V_{CB}$ = 10 - 30 V dc, $T_J$ = +150°C minimum. No heat sink or forced- air cooling on devices shall be permitted. n = 45, c = 0. |
| 2 | 1039 | The steady-state life test of step 1 shall be extended to 1,000 hours for each die design. Samples shall be selected from a wafer lot every twelve months of wafer production. Group B, step 2 shall not be required more than once for any single wafer lot. n = 45, c = 0. |
| 3 | 1032 | High- temperature life (non-operating), $T_A$ = +200°C, t = 340 hours, n = 22, c = 0. |

4.4.2.3 Group B sample selection. Samples selected from group B inspection shall meet all of the following requirements:

a. For JAN, JANTX and JANTXV samples shall be selected randomly from a minimum of three wafers (or from each wafer in the lot) from each wafer lot. For JANS, samples shall be selected from each inspection lot. See MIL-PRF-19500.

b. Must be chosen from an inspection lot that has been submitted to and passed group A, subgroup 2, conformance inspection. When the final lead finish is solder or any plating prone to oxidation at high temperature, the samples for life test (subgroups B4 and B5 for JANS, and group B for JAN, JANTX, and JANTXV) may be pulled prior to the application of final lead finish.

4.4.3 Group C inspection, Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table VII of MIL-PRF-19500, and in 4.4.3.1 (JANS).and 4.4.3.2 (JAN, JANTX, and JANTXV) herein for group C testing. Electrical measurements (end-points) and delta requirements shall be in accordance with group A, subgroup 2 and 4.5.5 herein delta parameters apply to subgroup C6.

4.4.3.1 Group C inspection, table VII (JANS) of MIL-PRF-19500.

| Subgroup | Method | Condition |
|---|---|---|
| C2 | 2036 | Test condition E, not applicable for U designation. |
| C6 | 1026 | 1,000 hours at $V_{CB}$ = 10 -30 V dc; $T_J$ = +150°C minimum. No heat sink or forced-air cooling on device shall be permitted. |

1/ Separate samples may be used for each step. In the event of a group B failure, the manufacturer may pull a new sample at double size from either the failed assembly lot or from another assembly lot from the same wafer lot. If the new “assembly lot” option is exercised, the failed assembly lot shall be scrapped.



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4.4.3.2 Group C inspection, table VII (JAN, JANTX, and JANTXV) of MIL-PRF-19500.

| Subgroup | Method | Condition |
|---|---|---|
| C2 | 2036 | Test condition E, not applicable for U designation. |
| C6 | | Not applicable. |

4.4.3.3 Group C sample selection. Samples for subgroups in group C shall be chosen at random from any inspection lot containing the intended package type and lead finish procured to the same specification which is submitted to and passes group A tests for conformance inspection. Testing of a subgroup using a single device type enclosed in the intended package type shall be considered as complying with the requirements for that subgroup.

4.4.4 Group E inspection. Group E inspection shall be performed for qualification or requalification only. The tests specified in table II herein must be performed to maintain qualification.

4.5 Method of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

4.5.2 Input capacitance. This test shall be conducted in accordance with method 3240 of MIL-STD-750, except the output capacitor shall be omitted.

4.5.3 Independent transistor inspections. Inspections shall be performed on each transistor in the array.

4.5.4 Transistor-to-transistor resistance. The leads of each transistor shall be shorted together for this test. The resistance shall be measured between each transistor in the array.

* 4.5.5 Delta requirements. Delta requirements shall be as specified below:

| Step | Inspection | MIL-STD-750 | | Symbol | Limit | Unit |
|---|---|---|---|---|---|---|
| | | Method | Conditions | | | |
| 1 | Collector-base cutoff current | 3036 | Bias condition D, $V_{CB}$ = 50 V dc | $\Delta I_{CB02}$ (1) | 100 percent of initial value or ± 8 nA dc, whichever is greater. | |
| 2 | Forward current transfer ratio | 3076 | $V_{CE}$ = 10 V dc; $I_C$ = 150 mA dc; pulsed see 4.5.1 | $\Delta h_{FE4}$ (1) | ±25 percent change from initial reading. | |

(1) Devices which exceed the group A limits for this test shall not be accepted.



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TABLE I. Group A inspection.

*

| Inspection 1/ | MIL-STD-750 | | Symbol | Limit | | Unit |
|---|---|---|---|---|---|---|
| | Method | Conditions | | Min | Max | |
| Subgroup 1 2/ | | | | | | |
| Visual and mechanical examination 3/ | 2071 | n = 45 devices, c = 0 | | | | |
| Solderability 3/ 4/ | 2026 | n = 15 leads, c = 0 | | | | |
| Resistance to solvents 3/ 4/ 5/ | 1022 | n = 15 devices, c = 0 | | | | |
| Temp cycling 3/ 4/ | 1051 | Test condition C, 25 cycles. n = 22 devices, c = 0 | | | | |
| Heremetic seal 4/<br>Fine leak<br>Gross leak | 1071 | n = 22 devices, c = 0 | | | | |
| Electrical measurements 4/ | | Group A, subgroup 2 | | | | |
| Bond strength 3/ 4/ | 2037 | Precondition $T_A$ = + 250°C at t = 24 hours or $T_A$ = + 300°C at t = 2 hours. n = 11 wires, c = 0 | | | | |
| Subgroup 2 | | | | | | |
| Collector to base, cutoff current | 3036 | $V_{CB}$ = 60 V dc | $I_{CBO1}$ | | 10 | µA dc |
| Emitter to base, cutoff current | 3061 | $V_{BE}$ = 5 V dc | $I_{EBO1}$ | | 10 | µA dc |
| Breakdown voltage collector to emitter | 3011 | Bias condition D; $I_C$ = 10 mA dc; pulsed (see 4.5.1) | $V_{(BR)CEO}$ | 60 | | V dc |
| Collector to base cutoff current | 3036 | Bias condition D; $V_{CB}$ = 50 V dc | $I_{CBO2}$ | | 10 | nA dc |
| Emitter to base cutoff current | 3061 | Bias condition D; $V_{EB}$ = 4 V dc | $I_{EBO2}$ | | 50 | nA dc |

See footnotes at end of table.



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TABLE I. Group A inspection - Continued.

*

| Inspection 1/ | MIL-STD-750 | | Symbol | Limit | | Unit |
|---|---|---|---|---|---|---|
| | Method | Conditions | | Min | Max | |
| Subgroup 2 - Continued | | | | | | |
| Forward-current transfer ratio | 3076 | $V_{CE}$ = 10 V dc; $I_C$ = 0.1 mA dc | $h_{FE1}$ | 75 | | |
| Forward-current transfer ratio | 3076 | $V_{CE}$ = 10 V dc; $I_C$ = 1.0 mA dc | $h_{FE2}$ | 100 | 450 | |
| Forward-current transfer ratio | 3076 | $V_{CE}$ = 10 V dc; $I_C$ = 10 mA dc | $h_{FE3}$ | 100 | | |
| Forward-current transfer ratio | 3076 | $V_{CE}$ = 10 V dc; $I_C$ = 150 mA dc pulsed (see 4.5.1) | $h_{FE4}$ | 100 | 300 | |
| Forward-current transfer ratio | 3076 | $V_{CE}$ = 10 V dc; $I_C$ = 500 mA dc pulsed (see 4.5.1) | $h_{FE5}$ | 50 | | |
| Collector to emitter voltage (saturated) | 3071 | $I_C$ = 150 mA dc; $I_B$ = 15 mA dc; pulsed (see 4.5.1) | $V_{CE(sat)1}$ | | 0.4 | V dc |
| Collector to emitter voltage (saturated) | 3071 | $I_C$ = 500 mA dc; $I_B$ = 50 mA dc; pulsed (see 4.5.1) | $V_{CE(sat)2}$ | | 1.6 | V dc |
| Base to emitter saturated voltage | 3066 | Test condition A; $I_C$ = 150 mA dc; $I_B$ = 15 mA dc; pulsed (see 4.5.1) | $V_{BE(sat)1}$ | | 1.3 | V dc |
| Base to emitter saturated voltage | 3066 | Test condition A; $I_C$ = 500 mA dc; $I_B$ = 50 mA dc; pulsed (see 4.5.1) | $V_{BE(sat)2}$ | | 2.6 | V dc |
| Subgroup 3 | | | | | | |
| High temperature operation: | | $T_A$ = +150°C | | | | |
| Collector to base cutoff current | 3036 | Bias condition D; $V_{CB}$ = 50 V dc | $I_{CBO3}$ | | 10 | µA dc |
| Low temperature operation: | | $T_A$ = -55°C | | | | |

See footnotes at end of table.



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TABLE I. Group A inspection - Continued.

*

| Inspection 1/ | MIL-STD-750 | | Symbol | Limit | | Unit |
|---|---|---|---|---|---|---|
| | Method | Conditions | | Min | Max | |
| Subgroup 3 – Continued | | | | | | |
| Forward-current transfer ratio | 3076 | $V_{CE}$ = 10 V dc; $I_C$ = 1.0 mA dc | $h_{FE6}$ | 50 | | |
| Subgroup 4 | | | | | | |
| Small-signal short-circuit forward-current transfer ratio | 3206 | $V_{CE}$ = 10 V dc; $I_C$ = 1 mA dc; f = 1 kHz | $h_{fe}$ | 100 | | |
| Magnitude of small-signal short-circuit forward- current transfer ratio | 3306 | $V_{CE}$ = 20 V dc; $I_C$ = 20 mA dc; f = 100 MHz | $\|h_{fe}\|$ | 2.0 | | |
| Open circuit output capacitance | 3236 | $V_{CB}$ = 10 V dc; $I_E$ = 0; 100 kHz ≤ f ≤ 1 MHz | $C_{obo}$ | | 8.0 | pF |
| Input capacitance (output open-circuited) | 3240 | $V_{EB}$ = 2.0 V dc; $I_C$ = 0; 100 kHz ≤ f ≤ 1 MHz; See 4.5.2 | $C_{ibo}$ | | 30 | pF |
| Turn-on time | | See figure 5 | $t_{on}$ | | 45 | ns |
| Turn-off time | | See figure 6 | $t_{off}$ | | 300 | ns |
| Transistor-to-transistor resistance | | $\|V_{T-T}\|$ = 500 V dc; see 4.5.4 | $R_{T-T}$ | $10^{10}$ | | ohms |
| Subgroups 5 and 6 | | | | | | |
| Not applicable | | | | | | |

See footnotes at end of table.



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TABLE I. Group A inspection - Continued.

*

| Inspection 1/ | MIL-STD-750 | | Symbol | Limit | | Unit |
|---|---|---|---|---|---|---|
| | Method | Conditions | | Min | Max | |
| Subgroups 7<br><br>Decap internal visual (design verification) | 2075 | n = 1 device, c = 0 | | | | |

1/ For sampling plan see MIL-PRF-19500.
2/ For resubmission of failed subgroup A1, double the sample size of the failed test or sequence of tests. A failure in group A, subgroup 1 shall not require retest of the entire subgroup. Only the failed test shall be rerun upon submission.
3/ Separate samples may be used.
4/ Not required for JANS devices.
5/ Not required for laser marked devices.



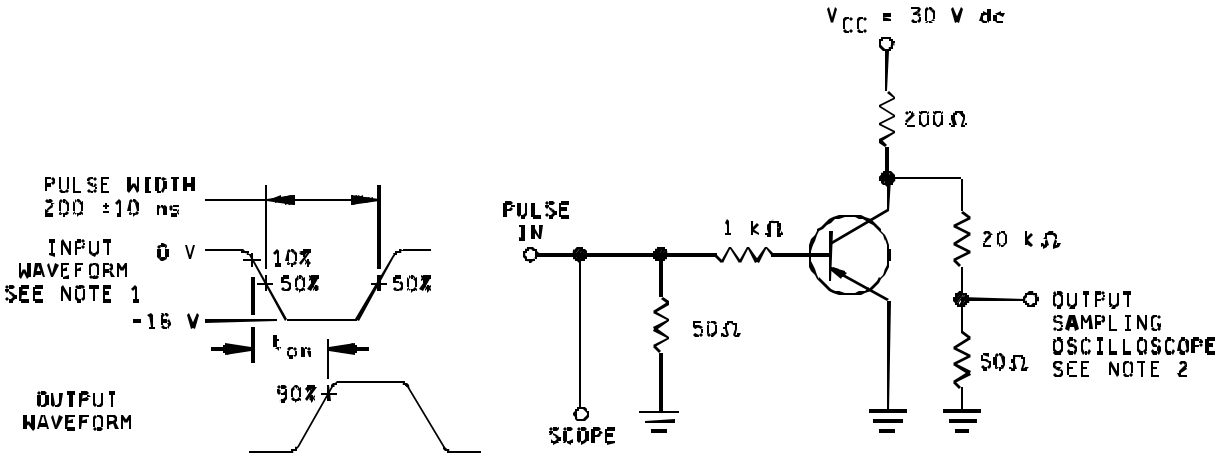
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* TABLE II. Group E inspection (all quality levels) - for qualification only.

| Inspection | MIL-STD-750 | | Qualification |
|---|---|---|---|
| | Method | Conditions | |
| Subgroup 1<br><br>Temperature cycling (air to air) | 1051 | Test condition C, 500 cycles | 45 devices<br>c = 0 |
| Hermetic seal<br><br>Fine leak<br>Gross leak | 1071 | | |
| Electrical measurements | | See group A, subgroup 2 and 4.5.5 herein. | |
| Subgroup 2<br><br>Intermittent life | 1037 | Intermittent operation life: $V_{CB}$ = 10 - 30 V dc , 6,000 cycles. | 45 devices<br>c = 0 |
| Electrical measurements | | See group A, subgroup 2 and 4.5.5 herein. | |
| Subgroup 3, 4, 5, 6 and 7<br><br>Not applicable | | | |
| Subgroup 8<br><br>Reverse stability | 1033 | Condition A for devices ≥ 400 V.<br>Condition B for devices < 400 V. | 45 devices<br>c = 0 |



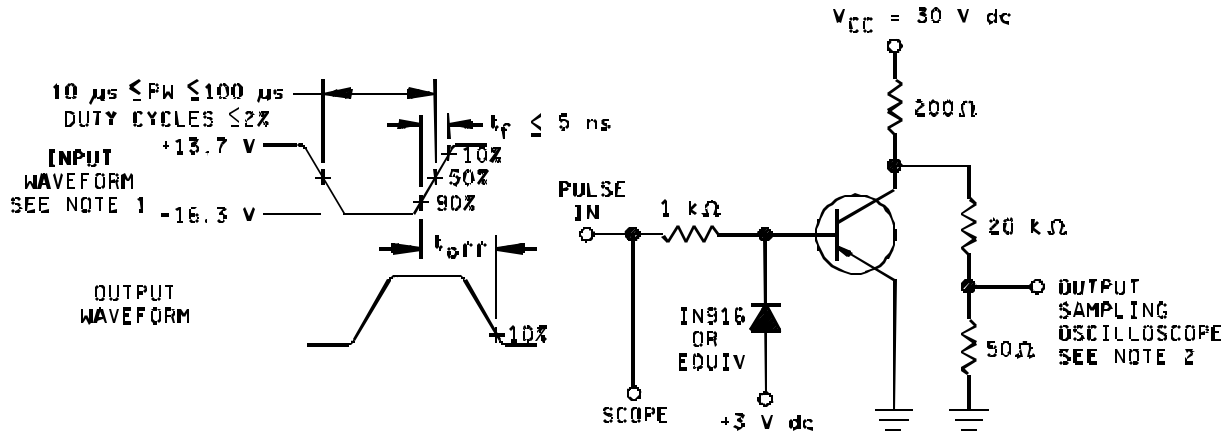
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NOTES:

1. The rise time ($t_r$) and fall time ($t_f$) of the applied pulse shall be each ≤ 2.0 ns; duty cycle ≤ 2 percent; generator source impedance shall be 50 Ω.
2. Output sampling oscilloscope: $Z_{in}$ ≥ 100 kΩ; $C_{in}$ ≤ 12 pF; rise time ≤ 5.0 ns.

FIGURE 5. Saturated turn-on switching time test circuit and waveform.




NOTES:

1. The rise time ($t_r$) and fall time ($t_f$) of the applied pulse shall be each ≤ 2.0 ns; duty cycle ≤ 2 percent; generator source impedance shall be 50 Ω.
2. Output sampling oscilloscope: $Z_{in}$ ≥ 100 kΩ; $C_{in}$ ≤ 12 pF; rise time ≤ 5.0 ns.

FIGURE 6. Saturated turn-off switching time test circuit and waveform.



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5. PACKAGING

5.1 Packaging. Packaging shall prevent mechanical damage of the devices during shipping and handling and shall not be detrimental to the device. When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Points' packaging activity within the Military Department or Defense Agency, or within the Military Departments' System Command. Packaging data retrieval is available from the managing Military Departments' or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The notes specified in MIL-PRF-19500 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.

b. Issue of DoDISS to be cited in the solicitation and if required, the specific issue of individual documents referenced (see 2.2.1).

c. The lead finish as specified (see 3.4.1).

d. Type designation and quality assurance level.

e. Packaging requirements (see 5.1).

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturer’s List (QML) No.19500 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Defense Supply Center, Columbus, ATTN: DSCC-VQE, P.O. Box 3990, Columbus, OH 43216-5000.

6.4 Changes from previous issue. The margins of this revision are marked with an asterisk to indicate where changes from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.



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Air Force - 11
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Review activities:
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Preparing activity:
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| **I RECOMMEND A CHANGE:** | 1. DOCUMENT NUMBER<br>MIL-PRF-19500/558D | 2. DOCUMENT DATE<br>14 January 2002 |
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3. **DOCUMENT TITLE**
SEMICONDUCTOR DEVICE, FOUR TRANSISTOR ARRAY,UNITIZED, PNP, SILICON, SWITCHING TYPES 2N6987, 2N6987U, AND 2N6988, JAN, JANTX, JANTXV, AND JANS

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

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| a. NAME (Last, First, Middle initial) | b. ORGANIZATION | |
|---|---|---|
| c. ADDRESS (Include Zip Code) | d. TELEPHONE (Include Area Code)<br>COMMERCIAL<br>DSN<br>FAX<br>EMAIL | 7. DATE SUBMITTED |

8. PREPARING ACTIVITY

| a. Point of Contact<br>Alan Barone | b. TELEPHONE<br>Commercial DSN FAX EMAIL<br>614-692-0510 850-0510 614-692-6939 alan.barone@dscc.dla.mil |
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